



Advanced Time Series Analysis and Forecasting

Seminar 10: Long memory and rough volatility



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BUX: Budapești Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CAC: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ CEE: Central and Eastern Europe
- ▣ DM: Diebold–Mariano (test)
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FCVAR: Fractionally Cointegrated VAR
- ▣ FFT: Fast Fourier Transform
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GPH: Geweke–Porter–Hudak (estimator)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ LW: Local Whittle (estimator)
- ▣ MSE: Mean Squared Error
- ▣ OLS: Ordinary Least Squares
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ QLIKE: Quasi-LIKelihood loss
- ▣ RFSV: Rough Fractional Stochastic Volatility (model)
- ▣ RV: Realised Variance

Acronyms used in this material (2/2)

- SD: **Standard Deviation**
- SE: **Standard Error**
- WIG20: **Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)**

Today's question and route

- **Question:** how do we measure, test and use long memory and roughness in volatility and inflation?
 - ▶ this seminar comes **before** Lecture 10: the section “What you need today” gives every definition the tasks use
- **Route**
 - ▶ Part A: ARFIMA autocorrelations, fractional weights, standard errors and bias of the estimators, fractional Gaussian noise, the RFSV kernel, circulant embedding, switching that mimics long memory
 - ▶ Part B: memory of FTSE 100 realised variance, BET and EUR/RON; roughness of the Euro Stoxx 50 and of Bitcoin; forecasting FTSE 100 and CAC 40 realised variance
 - ▶ Part C: co-memory of DAX and CAC 40 volatility; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	ARFIMA autocorrelations and the mean; fractional weights	Solved, Proposed	A1
A3, A4	standard errors of GPH and LW; bias and bandwidth	Solved, Proposed	A3
A5, A6	fractional Gaussian noise; the RFSV kernel	Solved, Proposed	A5
A7, A8	circulant embedding by hand; switching that mimics long memory	Solved, Proposed	A7, A1
B1, B2	memory of FTSE 100 log RV; BET and EUR/RON with the Qu test	Solved, Proposed	B1
B3, B4	roughness of the Euro Stoxx 50; Bitcoin by sampling interval	Solved, Proposed	B3
B5, B6	forecasting RV of the FTSE 100 and the CAC 40	Solved, Proposed	B5
C1, C2	fractional cointegration of two volatilities; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
RV of FTSE 100, Euro Stoxx 50, DAX, CAC 40	Heber et al. (2009)	daily, 5-minute returns	2000–2022
RV of Bitcoin	Binance one-minute prices	daily, 1–60 minutes	2018–2026
BET	EODHD	daily	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- RV in $\%^2$, returns in %; in the notebook: `gph`, `local_whittle`, `elw`, `qu_stat`, `gjr_scaling`, `h_with_noise`, `rfsv_kernel`, `forecast_rv`, `fcvar_fit`; no account or key is needed

What you need today (1/4): long memory

- Long memory: $\rho(k) \sim c k^{2d-1}$ and $f(\lambda) \sim G\lambda^{-2d}$ near zero, $0 < d < 1/2$; Hurst exponent $H = d + 1/2$
 - ▶ ARFIMA(0, d , 0): $(1 - L)^d X_t = \varepsilon_t$; $(1 - L)^d = \sum_k \pi_k L^k$, $\pi_0 = 1$, $\pi_k = \pi_{k-1}(k - 1 - d)/k$ [Hosking \(1981\)](#)
 - ▶ $\rho(k) = \rho(k - 1)(k - 1 + d)/(k - d)$, $\rho(k) \sim \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$; $\pi_k \sim k^{-d-1}/\Gamma(-d)$
- Variance of the mean: $\text{Var}(\bar{X}_n) = \frac{1}{n} \sum_{|k| < n} (1 - |k|/n) \gamma(k)$, of order n^{2d-1} , not n^{-1}
- Periodogram $I(\lambda_j) = \frac{1}{2\pi n} |\sum_t X_t e^{-i\lambda_j t}|^2$ at Fourier frequencies $\lambda_j = 2\pi j/n$; bandwidth m : the number of low frequencies used

What you need today (2/4): estimators of d

- ▣ **GPH** Geweke and Porter-Hudak (1983): OLS of $\log I(\lambda_j)$ on $-\log(4 \sin^2(\lambda_j/2))$, $j \leq m$; SE $\pi/\sqrt{24m}$
- ▣ **Local Whittle** Robinson (1995b): $\hat{d} = \arg \min_d \log \left(\frac{1}{m} \sum_j \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_j \log \lambda_j$; SE $1/(2\sqrt{m})$
 - ▶ **exact local Whittle** Shimotsu and Phillips (2005), Shimotsu (2010): the periodogram of $(1-L)^d(X_t - \hat{\mu})$ instead of $\lambda_j^{2d} I(\lambda_j)$; valid also for $d \geq 1$
- ▣ Bias for ARFIMA(1, d , 0): $\mathbb{E}\hat{d} - d \approx C(m/n)^2$, $C = \frac{2\pi^2}{9} \cdot \frac{2\phi}{(1-\phi)^2}$; MSE-optimal $m^* = (n^4/(16C^2))^{1/5}$

What you need today (3/4): breaks and co-memory

- Level shifts, breaks and rare regime switches create a spectral pole: $\hat{d}$ is large although the process has short memory [Diebold and Inoue \(2001\)](#)
 - ▶ **Qu test** [Qu \(2011\)](#): $W = \sup_{r \in [\varepsilon, 1]} (\sum_j \nu_j^2)^{-1/2} |\sum_{j \leq mr} \nu_j (\lambda_j^{2\hat{d}} I(\lambda_j) / \hat{G} - 1)|$, $\nu_j = \log \lambda_j - \overline{\log \lambda}$; reject stationary long memory for large W
 - ▶ $m = n^{0.7}$, $\varepsilon = 0.02$; simulated critical values (10%, 5%, 1%): 1.09, 1.22, 1.56
- **FCVAR** [Johansen and Nielsen \(2012\)](#): $\Delta^d X_t = \alpha \beta' L_b \Delta^{d-b} X_t + \varepsilon_t$, $L_b = 1 - \Delta^b$; $\beta' X_t$ has memory $d - b$
 - ▶ for fixed (d, b) : Johansen reduced-rank regression; trace tests $\chi_{(p-r)^2}^2$ when $b < 1/2$

What you need today (4/4): rough volatility

- fBm W^H : $\mathbb{E}|W_{t+\Delta}^H - W_t^H|^2 = \Delta^{2H}$; increments (fGn) $\rho(k) = \frac{1}{2}(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H})$ Mandelbrot and Van Ness (1968)
 - circulant embedding Davies and Harte (1987): first row $(\gamma_0, \dots, \gamma_{n-1}, \gamma_{n-2}, \dots, \gamma_1)$, eigenvalues = FFT of the row; exact if all ≥ 0
- Gatheral, Jaisson and Rosenbaum (2018): $m(q, \Delta) = \text{mean} |\log \sigma_{t+\Delta} - \log \sigma_t|^q \propto \Delta^{\zeta_q}$, $\zeta_q = qH$, $\sigma_t = \sqrt{RV_t}$; with i.i.d. error: $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$
- RFSV predictor: $\widehat{\log \sigma_{t+h}^2} = \sum_k w_k \log \sigma_{t-k}^2$, $w_k \propto \int_k^{k+1} \frac{du}{(u+h)u^{H+1/2}}$; variance: $\exp(\cdot + 2c\nu^2 h^{2H})$,
 $c = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$
 - QLIKE loss $RV/F - \log(RV/F) - 1$ Patton (2011); DM test with Newey–West variance (Chapter 1)

A1: ARFIMA autocorrelations and the mean [Solved]

Task

- ▣ ARFIMA(0, 0.3, 0).
- ▣ 1. Show that $\rho(1) = d/(1 - d)$ and compute $\rho(1)$, $\rho(10)$, $\rho(100)$ with the recursion.
- ▣ 2. Compare $\rho(100)$ with $\frac{\Gamma(1-d)}{\Gamma(d)} 100^{2d-1}$.
- ▣ 3. For $n = 1000$, compute the ratio of the true SD of $\bar{X}$ to the i.i.d. formula $\sqrt{\gamma(0)/n}$ (notebook A1).
- ▣ Report: three numbers; two numbers; one number and one sentence.

Solution

- ▣ 1. $\rho(1) = \rho(0) \cdot \frac{0+d}{1-d}$; $\rho(1) = 0.429$, $\rho(10) = 0.173$, $\rho(100) = 0.069$
- ▣ 2. Asymptotic formula: 0.069, practically equal already at $k = 100$
- ▣ 3. Ratio 7.55: the i.i.d. standard error is 7.55 times too small
- ▣ Interpretation: with $d = 0.3$, 1000 observations carry the information of far fewer independent ones; HAC with a fixed bandwidth cannot repair this

A2: fractional differencing weights [Proposed]

Solution discussed in class

Task

- ▣ $d = 0.4$. Model: A1.
- ▣ 1. Compute $\pi_1, \dots, \pi_5$ of $(1 - L)^{0.4}$.
- ▣ 2. Compare π_{100} with $100^{-d-1}/\Gamma(-d)$.
- ▣ 3. Compute $\sum_{k=0}^{1000} \pi_k$; what is its limit, and what does the gap mean for a truncated filter?
- ▣ Report: five numbers; two numbers; one number and one sentence.

A3: standard errors of GPH and local Whittle [Solved]

Task

- ▣ $n = 5000$, $m = \lfloor n^{0.65} \rfloor$; a local Whittle estimate $\hat{d} = 0.45$.
- ▣ 1. Compute m and the asymptotic SE of GPH and of local Whittle.
- ▣ 2. Give the 95% interval for d and the relative efficiency of LW against GPH.
- ▣ 3. Which m gives a half-width of 0.05?
- ▣ Report: three numbers; an interval and one number; one number.

Solution

- ▣ 1. $m = 253$; GPH $\pi/\sqrt{24m} = 0.0403$; LW $1/(2\sqrt{m}) = 0.0314$
- ▣ 2. $[0.388, 0.512]$; efficiency $(\pi^2/24)/(1/4) = \pi^2/6 = 1.645$
- ▣ 3. $1.96/(2\sqrt{m}) \leq 0.05 \Rightarrow m \geq 385$
- ▣ Interpretation: the interval includes $1/2$, so stationarity is not established; precision is bought with bandwidth, and bandwidth costs bias (A4)

A4: bias and the optimal bandwidth [Proposed]

Solution discussed in class

Task

- ▣ ARFIMA(1, d , 0) with $\phi = 0.5$, $n = 5000$. Model: A3.
- ▣ 1. Compute the bias constant C .
- ▣ 2. For $m = \lfloor n^{0.8} \rfloor$, compute the bias and the SD of local Whittle.
- ▣ 3. Compute m^* , its exponent $\log m^* / \log n$, and the bias and SD at m^* .
- ▣ Report: one number; two numbers; four numbers and one sentence.

A5: fractional Gaussian noise and scaling [Solved]

Task

- ▣ fGn with $H = 0.1, 0.3$ and 0.7 .
- ▣ 1. Compute $\rho(1) = 2^{2H-1} - 1$ and $\rho(2)$ for each H .
- ▣ 2. Compute $m(2, 22)/m(2, 1) = 22^{2H}$: how much more does log volatility move over a month than over a day?
- ▣ 3. Show that $\zeta_q = qH$ for fBm.
- ▣ Report: six numbers; three numbers; one derivation.

Solution

- ▣ 1. $H = 0.1$: $-0.426, -0.026$; $H = 0.3$: $-0.242, -0.049$; $H = 0.7$: $0.320, 0.189$
- ▣ 2. $1.86, 6.39, 75.75$: with $H = 0.1$ the monthly variance of changes is less than twice the daily one
- ▣ 3. $W_{t+\Delta}^H - W_t^H \sim N(0, \Delta^{2H})$: $\mathbb{E}|\cdot|^q = \Delta^{qH} \mathbb{E}|Z|^q$; take logs
- ▣ Interpretation: rough log volatility moves a lot from day to day but drifts little: strong negative correlation of daily changes

A6: the RFSV kernel and the variance correction [Proposed]

Solution discussed in class

Task

- $H = 0.1$, $\nu = 0.3$, horizon $h = 1$; kernel truncated at 500 days. Model: A5.
- 1. Compute the share of the weight on the last day, the last 5 and the last 22 days (notebook A6).
- 2. Compute $c(H)$ and the correction $2c\nu^2 h^{2H}$ for $h = 1, 5, 22$.
- 3. By how much would ignoring the correction understate the variance forecast at $h = 22$?
- Report: three shares; four numbers; one number.

A7: circulant embedding by hand [Solved]

Task

- ▣ Three points of fGn with $H = 0.3$: $\gamma_0, \gamma_1, \gamma_2$.
- ▣ 1. Compute γ_1 and γ_2 .
- ▣ 2. Write the first row of the 4×4 circulant and its eigenvalues $\lambda_k = \sum_j c_j e^{-2\pi ijk/4}$.
- ▣ 3. Can the Davies–Harte method be used?
- ▣ Report: two numbers; a row and four numbers; one sentence.

Solution

- ▣ 1. $\gamma_1 = \frac{1}{2}(2^{0.6} - 2) = -0.2421$,
 $\gamma_2 = \frac{1}{2}(3^{0.6} - 2 \cdot 2^{0.6} + 1) = -0.0491$
- ▣ 2. $c = (\gamma_0, \gamma_1, \gamma_2, \gamma_1)$; $\lambda_0 = \gamma_0 + 2\gamma_1 + \gamma_2 = 0.4666$,
 $\lambda_1 = \lambda_3 = \gamma_0 - \gamma_2 = 1.0491$, $\lambda_2 = \gamma_0 - 2\gamma_1 + \gamma_2 = 1.4352$
- ▣ 3. Yes: all eigenvalues are positive, so the circulant is a valid covariance and its first 3 coordinates have the fGn covariance
- ▣ Interpretation: the embedding doubles the size but turns an $O(n^3)$ factorisation into two FFTs

A8: switching that mimics long memory [Proposed]

Solution discussed in class

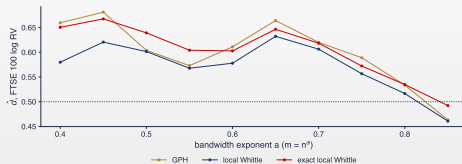
Task

- ▣ $X_t = \mu S_t + \varepsilon_t$, $S_t \in \{-1, 1\}$ a Markov chain that stays with probability $q = 0.995$, $\mu = 1$, $\varepsilon_t \sim N(0, 1)$. Model: A1.
- ▣ 1. Show that $\rho(k) = \frac{\mu^2}{\mu^2 + 1} (2q - 1)^k$ and compute $\rho(1)$, $\rho(100)$, $\rho(500)$.
- ▣ 2. Compute the half-life of the autocorrelation.
- ▣ 3. Fit $\log \rho(k)$ on $\log k$ for $k = 5, \dots, 200$ and convert the slope to d .
- ▣ Report: three numbers; one number; one number and one sentence.

B1: memory of FTSE 100 realised variance [Solved]

- ▣ **Question:** how large is d for the log realised variance of the FTSE 100, and is it robust to the bandwidth and to level shifts?
- ▣ **Inputs:** Oxford-Man 5-minute RV of the FTSE 100, $n = 5\,586$ days to 25 February 2022
- ▣ **Tasks**
 1. Estimate d of log RV by GPH, local Whittle and exact local Whittle for $m = n^{0.5}, n^{0.6}, n^{0.7}$, with standard errors.
 2. Plot $\hat{d}(m)$ for $m = n^{0.40}, \dots, n^{0.85}$.
 3. Compute the Qu statistic with $m = n^{0.7}$ and $\varepsilon = 0.02$.
 4. Interpretation: is the memory of FTSE 100 volatility genuine, and is it stationary?
- ▣ **Report:** a table of nine estimates with SEs, one chart, one statistic, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



m	GPH (SE)	LW	ELW	SE (LW, ELW)
74	0.604 (0.075)	0.601	0.639	0.058
177	0.611 (0.048)	0.578	0.603	0.038
419	0.620 (0.031)	0.606	0.618	0.024

- Qu: $W = 0.33$ against the 10% value 1.09: no evidence of level shifts
- Interpretation: $\hat{d} \approx 0.6$ at every bandwidth with a flat $\hat{d}(m)$: genuine long memory, at or just beyond the stationarity border; report ELW

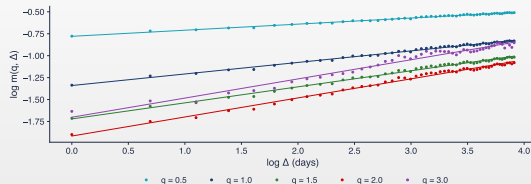
B2: BET and EUR/RON with the Qu test [Proposed]

- **Question:** is the memory of $|r_t|$ on the Bucharest market and for the leu genuine, or does it come from regime changes?
- **Inputs:** BET 2000–2026 and from 2010; EUR/RON (BNR) 2005–2026 and from 2014; model: B1
- **Tasks**
 1. Estimate d of $|r_t|$ by local Whittle with $m = n^{0.65}$ on the full sample and after the cut date.
 2. Compute the Qu statistic ($m = n^{0.7}$, $\varepsilon = 0.02$) on the same samples.
 3. Interpretation: which of the two markets has genuine long memory in volatility?
- **Report:** four estimates with SEs, four statistics, one sentence
- **Notebook:** section B2

B3: roughness of the Euro Stoxx 50 [Solved]

- ▣ **Question:** does the Gatheral–Jaisson–Rosenbaum scaling hold for the Euro Stoxx 50, and how rough is its volatility?
- ▣ **Inputs:** Oxford-Man 5-minute RV, $n = 5\,635$ days; $\sigma_t = \sqrt{RV_t}$; $q = 0.5, 1, 1.5, 2, 3$; $\Delta = 1, \dots, 50$
- ▣ **Tasks**
 1. Compute $m(q, \Delta)$ and the slopes ζ_q ; estimate H by regressing ζ_q on q through the origin.
 2. Estimate H with the measurement-error intercept $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$.
 3. Repeat the first estimate on the two halves of the sample.
 4. Interpretation: is ζ_q linear in q , and what does a departure mean?
- ▣ **Report:** five slopes, three values of H , ν , one sentence
- ▣ **Notebook:** section B3

B3: solution [Solved]



- $\zeta_q = 0.069, 0.132, 0.183, 0.217, 0.218$; $\hat{H} = 0.093$, $\hat{\nu} = 0.38$; with the error intercept 0.109; halves 0.091 and 0.095
- Interpretation: linear up to $q = 2$, flat between $q = 2$ and 3: large moments are driven by a few extreme days (heavy tails of the increments), so H from $q \leq 2$ is the safer reading; roughness is stable over time

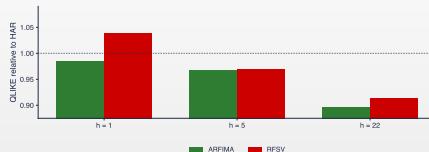
B4: Bitcoin roughness by sampling interval [Proposed]

- **Question:** does the estimate of H for Bitcoin depend on the sampling interval of the realised variance?
- **Inputs:** Binance RV of Bitcoin from 1-, 5-, 15-, 30- and 60-minute returns, 2018–2026; model: B3
- **Tasks**
 1. Estimate H for each RV by the moment regression and with the error intercept.
 2. Estimate d of $\log RV$ by local Whittle with $m = n^{0.65}$.
 3. Interpretation: which direction does a coarser grid push H , and why?
- **Report:** ten values of H , five values of d , one sentence
- **Notebook:** section B4

B5: forecasting FTSE 100 realised variance [Solved]

- **Question:** does a rough or a long-memory model forecast FTSE 100 realised variance better than HAR?
- **Inputs:** rolling window of 1000 days, re-estimation every 20 days; out of sample 30 December 2003 – 26 January 2022 ($T = 4\,565$); horizons 1, 5, 22 days
- **Tasks**
 1. Forecast RV_{t+h} with HAR, ARFIMA(0, d , 0) and RFSV, with the lognormal corrections.
 2. Compute the average QLIKE of each model and the ratio to HAR.
 3. Compute the DM statistics against HAR.
 4. Interpretation: which model would you use at each horizon?
- **Report:** nine losses, six ratios, six DM statistics, one sentence
- **Notebook:** section B5

B5: solution [Solved]



h	QLIKE HAR	ARFIMA / HAR	DM	RFSV / HAR	DM
1	0.230	0.985	-0.98	1.038	1.49
5	0.324	0.968	-1.94	0.968	-1.33
22	0.561	0.895	-2.12	0.914	-1.93

- Interpretation: HAR at one day (RFSV is worse); at 22 days both ARFIMA ($p = 0.034$) and RFSV ($p = 0.053$) beat HAR by about a tenth; median rolling $\hat{H} = 0.11$

B6: forecasting CAC 40 realised variance [Proposed]

- **Question:** does the FTSE 100 ranking carry over to the CAC 40?
- **Inputs:** the same design, out of sample 22 December 2003 – 26 January 2022 ($T = 4\,631$); model: B5
- **Tasks**
 1. Repeat the three forecasts and the QLIKE ratios.
 2. Compute the DM statistics against HAR.
 3. Interpretation: is there a single best model across the two indices?
- **Report:** six ratios, six DM statistics, one sentence
- **Notebook:** section B6

C1: co-memory of DAX and CAC 40 volatility [Proposed]

- ▣ **Question:** do the volatilities of two neighbouring markets share one fractional trend?
- ▣ **Inputs:** log 5-minute RV of the DAX and the CAC 40 (Oxford-Man), $n = 5\,600$ common days; FCVAR with $k = 0$; model: B1
- ▣ **Tasks**
 1. Write a five-line pre-registration: hypothesis, model, grid of (d, b) , tests, what would change your mind.
 2. Estimate d of each series and of the spread $\beta'X_t$ by local Whittle, then the FCVAR by profile likelihood with trace tests.
 3. Interpretation: what do the data support, and what would you add to the model?
- ▣ **Report:** a five-line plan, five estimates, two tests, a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise long memory and rough volatility. The answer:
- ▣ (a) An ARFIMA with $d = 0.3$ has autocorrelations that decay exponentially, only more slowly than an AR(1).
- ▣ (b) The standard error of local Whittle is $\pi/\sqrt{24m}$.
- ▣ (c) A larger bandwidth m is always better, because the variance falls.
- ▣ (d) If the Qu test rejects, the series has structural breaks.
- ▣ (e) $H = 0.1$ means that volatility has short memory, because $H < 0.5$ is antipersistence.
- ▣ (f) FIGARCH with $0 < d < 1$ is covariance stationary with a finite unconditional variance.
- ▣ (g) Exact local Whittle remains consistent for $d = 1.2$, whereas local Whittle does not.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Long memory makes i.i.d. and HAC standard errors far too small
- ▣ Local Whittle is the default; exact local Whittle when d may exceed $1/2$; always vary m
- ▣ Test against level shifts before claiming long memory
- ▣ Log volatility is rough ($H \approx 0.1$) and persistent; the proxy affects H
- ▣ Forecast gains over HAR are small and horizon-dependent

After the seminar

- ▣ Lecture 10 develops each topic of today: spectral theory, asymptotics of the estimators, the Qu test, FCVAR, FIGARCH, rough volatility and its forecasts
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: co-memory of Central and Eastern European volatilities
- ▣ Reading: [Robinson \(1995b\)](#); [Shimotsu and Phillips \(2005\)](#); [Qu \(2011\)](#); [Gatheral, Jaisson and Rosenbaum \(2018\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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